

TECT epoch 7 retrospective (Math70–84, 2026-04-22 to 04-23): Pillars 4–7 advancement and first stage-wide scorecard snapshot

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This paper is an EPOCH RETROSPECTIVE compiling the internal theorem programme established by the Math70–84 archive of the Topological Energy Condensate Theory (TECT) during 2026-04-22 to 04-23. This epoch consolidates the then-current advancement of Pillars 4–7 and records the first stage-wide scorecard, without claiming that the later Math270–313 defence and verdict cycles were already complete. The four advancement themes recorded in the Math70–84 archive were: candidate Pillar 4 gauge-group emergence via defect-bundle topology on $\mathbb{CP}^2$ and equivariant-moduli arguments; candidate Pillar 5 chirality consolidation via index-theoretic protection; candidate Pillar 6 generational structure via a Pati–Salam RGE + Yukawa hierarchy; and candidate Pillar 7 quantum consistency via Ward-identity and modular-arithmetic checks. The first stage-wide tier-coherent snapshot was a then-current 4 PROVED (T7) / 2 PROVED CONDITIONAL (T6) / 1 CLOSED@1-LOOP (T5) / 3 STRONG EVIDENCE (T4) / 1 OPEN-NEGATIVE (T0+T2) summary; the current canonical scorecard is recorded in `Docs/status/TOE-FACT-SHEET.md` and reflects the Math270–313 defence cycles and the Math302 proof-rigor / realisation-gate carve-out. This document does NOT advance the canonical tier.

I. INTRODUCTION

The TECT framework rests on an 11-pillar foundation (Math54), each addressing a distinct sector of fundamental physics: emergence of the mass gap, Lorentz invariance, spin-2 gravity, electroweak gauge symmetry, chirality, generations, quantum consistency, transversality, equivalence principle, $\hbar$ -origin, and the cosmological constant. After four Epochs of foundational work (Math01-69), the research programme required a systematic tier assignment reflecting the current mathematical status of each pillar. Epoch 7 delivers that assessment and simultaneously advances four pillars to new closure depths.

This paper documents the chronological arc of 2026-04-22 to 04-23, chronicling (i) the Pillar 4 gauge-group programme under Math75 + Math80 (defect moduli, equivariance, $SO(10)$ uniqueness,

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Yukawa hierarchy), (ii) the Pillar 5 chirality closure via Math76 (index-theoretic protection), (iii) the Pillar 6 generational embedding via Math77 (Pati-Salam + RGE), (iv) the Pillar 7 quantum consistency via Ward + modular constraints, and (v) the integration of all 11 pillars into a canonical Stage-1 scorecard (Math78 synthesis, later audit-rolled-back to PARTIAL-ADVANCED but directionally sound).

II. SETTING AND PILLAR HIERARCHY

The TECT 11-pillar structure (Math54) partitions the theoretical scope as:

- **Pillar 1:** Mass gap m^* (Brazovskii-regime locked branch).
- **Pillar 2:** IR-scale Lorentz invariance (kinematic bound + shell isotropy).
- **Pillar 3:** Emergent spin-2 gravity (one-loop via Ward identity).
- **Pillar 4:** Electroweak + GUT gauge group emergence and uniqueness.
- **Pillar 5:** Chirality from index-theoretic protection (Dirac fermion structure).
- **Pillar 6:** Generational structure and SM embedding (Higgs+three-family).
- **Pillar 7:** Per-generation quantum consistency (Ward + modular arithmetic).
- **Pillar 8:** Lorentz interval-certificate transversality.
- **Pillar 9:** Equivalence principle from metric perturbation dynamics.
- **Pillar 10:** Planck constant $\hbar$ as cosmic phase-transition adiabatic invariant.
- **Pillar 11:** Cosmological constant Λ via four-sector cancellation.

Epoch 7 focuses on Pillars 4-7, each requiring independent mathematical closure at the proof-sketch (T3) or strong-evidence (T4) level or above before Stage-1 scorecard publication.

III. MATHEMATICAL CONTRIBUTIONS

a. Pillar 4: Gauge-group emergence (Math75, Math80, Math80-AddA-D, Math80-AddE)
Math75 initiates the Pillar 4 closure programme, addressing three sub-questions (Q1 equivariant cohomology, Q2 RG-flow derivation, Q3 symplectic reduction) via the order-parameter structure

on the BCC defect manifold. The work is PROVED CONDITIONAL (T6) on the assumption that the Grassmannian quotient $\text{Gr}(2, 5)$ with $\text{SU}(5)$ acting via the two-dimensional representation yields the Standard Model gauge group via stabiliser theorem. Math80 (strategy framework) outlines the full gauge-emergence research programme, including defect-bundle topology on $\mathbb{CP}^2$, equivariant-moduli space structure, and Yukawa coupling emergence from RGE boundary conditions.

Math80-AddA (defect-moduli theorem) establishes that the two-dimensional $\text{SU}(5)$ representation moduli on $\text{Gr}(2, 5)$ parametrises the six $\text{U}(1)$ charges of the SM. The result is PROVED (T7): all six charge assignments are derivable from equivariant-action considerations without external input. Math80-AddB (equivariance closure) proves that the $\text{U}(1)_Y$ charge distribution respects the required equivariance under the full gauge group, linking topological and representation-theoretic constraints. Math80-AddC ($\text{SO}(10)$ uniqueness) establishes $\text{SO}(10)$ as the unique GUT containing the SM gauge group via the four-flavour Yukawa structure, PROVED CONDITIONAL (T6) on the single-shell approximation assumption (caveat documented). Math80-AddE (Yukawa hierarchy) shows that the hierarchy $y_t > y_b > y_\tau$ emerges from RGE scaling of the BCC elastic modulus across the Margnon transition scale, PROVED (T7).

Verdict on Pillar 4: T6 PROVED CONDITIONAL on defect-bundle transitivity.

b. Pillar 5: Chirality from index-theoretic protection (Math76) Math76 records the then-current Pillar 5 closure programme: the three-dimensional Dirac-node structure at the BCC-to-condensate phase boundary carries a topological invariant (the spectral flow of the associated Clifford operator) that protects the chirality of the fermionic zero mode. At the time of writing of this epoch, the result was tagged PROVED (T7) on the basis of Math10–14 (Pillar 5 five-layer mechanism) plus the Math76 boundary-operator index-theoretic argument. Subsequent canonical bookkeeping records the chirality emergence at T7 PROVED with the 16-zero-mode count HRR-formula-conditional per Math171-AddA / Top-impact paper TI-1. The three-dimensional Dirac-node argument employs Atiyah–Singer index methods applied to the boundary operator D_∂ and the two-valley structure in the order-parameter manifold; the argument is recorded here as the then-current closure narrative, not as a re-derivation in the present paper.

Then-current epoch-7 verdict on Pillar 5: T7 PROVED (chirality emergence). Current canonical tier per Docs/status/TOE-FACT-SHEET.md: T7 PROVED with the HRR-conditional zero-mode count tracked separately.

c. Pillar 6: Generational structure (Math77) Math77 addresses the generational structure of the SM within TECT via a two-step RGE flow: (i) $\text{SU}(5)$ GUT scale breakdown to SM via Pati-Salam intermediate structure; (ii) family-structure emergence from three-component order-

parameter valleys. The result is PROVED CONDITIONAL (T6) on the Pati-Salam intermediate symmetry assumption (standard in GUT literature but introduced here as an explicit hypothesis). The Yukawa coupling hierarchy $y_t \gg y_b \gg y_\tau$ is derived from RGE running, not postulated.

Verdict on Pillar 6: T6 PROVED CONDITIONAL.

d. Pillar 7: Quantum consistency (Math47, Math48 carryover + modular constraints) Pillar 7 closure combines Math47-Math48 Ward identity closure (carried over from Epoch 5) with explicit modular-arithmetic constraints on the perturbed metric $h^{\mu\nu}$ ensuring that quantum-state-count calculations remain non-negative modulo the perturbative order. The constraint encodes the requirement that the full gravitational backreaction does not produce negative probabilities at any order. Verdict: T4 STRONG EVIDENCE (proof sketch complete; full analytical verification multi-turn project, currently partial-automated in the Math82-H continuum-audit framework).

e. Stage-1 Scorecard Synthesis (Math78, audit-rolled-back) Math78 attempts a global stage-wide tier assignment for all 11 pillars, concluding that 4 pillars reach T7 (PROVED), 2 reach T6 (PROVED CONDITIONAL), 1 reaches T5 (CLOSED@1-LOOP), 3 reach T4 (STRONG EVIDENCE), and 1 is open-negatively-defined (Pillar 10, $\hbar$ origin, provisionally unsolved in classical framework, leading to the quantum-sector programme). The scorecard is later (post-Epoch 8) audit-rolled back from "Stage-1 complete" to "Stage-1 PARTIAL-ADVANCED" due to discovery of hidden dependencies between pillars (see Epoch 8-9 narrative), but the tier-assignment methodology itself is sound and forms the basis for the refined Stage-2 gate programme.

IV. TIER CLASSIFICATION AND AUDIT TRAIL

This epoch establishes TECT's first canonical tier-assignment framework (see the internal status-tier nomenclature policy for full tier semantics; binding from 2026-04-29):

- **T7 (PROVED):** Pillar 5 (Math76 chirality), Pillar 3 carryover (gravity), Pillars 1 & 8 carryover (mass gap, transversality).
- **T6 (PROVED CONDITIONAL):** Pillar 4 (Math80-AddA/C, SO(10) uniqueness on single-shell), Pillar 6 (Math77, Pati-Salam intermediate stage).
- **T5 (CLOSED@1-LOOP):** Pillar 3 loop-order explicit.
- **T4 (STRONG EVIDENCE):** Pillar 7 (Math47-48 + modular audit), Pillar 9 (equivalence principle), Pillar 11 (cosmological constant four-sector).

- **T2 (CONJECTURE):** Pillar 10 ($\hbar$ classical no-go, opens quantum programme).

The scorecard assignment is the canonical reference tier-state as of 2026-04-23. Subsequent epochs (8-12) refine this through further audit and dependent-structure discovery.

V. DISCUSSION AND OUTLOOK

Epoch 7 represents the systematic maturation of TECT’s proof-tier methodology. The 11-pillar structure, previously treated as a conceptual checklist, now becomes a formal tier-assignment matrix, enabling quantitative tracking of progress toward Stage-1 closure (overall tier = minimum of component tiers). Stage-1, as defined in Math60, requires all 11 pillars to reach at least T3 (proof sketch); Epoch 7 demonstrates that 8 pillars already exceed T3, with only Pillar 10 requiring escalation from conjecture to proof-sketch level.

The Pillar 4 closure programme (Math75, Math80) sets the tone for Epochs 8-12: mathematically deep, making few assumptions, and traceable to numerical evidence where applicable. The defect-moduli theorem (Math80-AddA) is particularly notable as the first purely index-theoretic gauge-group derivation in TECT, requiring no external Higgs sector postulate.

Looking ahead, Epochs 8-9 will pivot toward the quantum sector: $\hbar$ -origin (Pillar 10), quantum-completion gates (GAP-1 through GAP-4), and the integration of Stage-2 closure conditions (Math60 sub-theorems). The Pillar 4 atlas programme (Epoch 9, Math162-167) will deepen the defect-bundle structure initiated here, potentially upgrading Pillar 4 from T6 to T7 if the three-patch Čech-cocycle closure passes audit.

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